

Amar Dhillon

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EDUCATION

University of Maryland, College Park

B.S. in Computer Science, Minor in Statistics | GPA: 3.5

Courses: Object Oriented Programming I/II, Algorithms I/II, Computer Systems, Artificial Intelligence, Data Science, Applied Statistics I/II

College Park, MD

May 2028

CERTIFICATIONS

Amazon Web Services (AWS) Certified Developer - Associate

Amazon Web Services

Nov 2024 – Nov 2027

Digital Badge

EXPERIENCE

Software Development Engineer Intern

Amazon Web Services

Sep 2026 – Dec 2026

Redmond, WA

- Researching and implementing **Java**-based utilization optimizations for **Amazon Redshift**

Software Engineering Intern

Capital One

June 2026 – August 2026

Richmond, VA

- Overhauled the legacy UI of a test data workflow creator using **React.js** + **Electron** with **TypeScript**-aided workflow and state validation backed by **Jest** testing on mock workflows and improved editor fluidity via **React-Flow**, receiving an internal satisfaction report of **90%** compared to the legacy tool
- Engineered a lightweight tree-style workflow search index to minimize database queries under access constraints, implementing a **Python** + **SQLAlchemy** pipeline that distilled popular workflows from a **270,000+** row **PostgreSQL** table into a precomputed search index, backed by **Redis** caching for full workflow retrieval and abstraction through a **Node.js** IPC module
- Developed a **Claude** skill for workflow generation, constraining open-ended model search via **sub-skill decomposition** and **Python** functions for workflow search and validation, and used **AWS CloudWatch** + **OpenSearch** latency logging to locate bottlenecks, cutting generation time **65%** from previous tooling to **under 15 seconds**

Software Engineer (Contract)

Amazon (via UMD App Development Contracting)

Jan 2026 – Present

College Park, MD

- Deployed an **Amazon Leo** space topic research utility adopted into Leo's ecosystem through **Docker** + **AWS Fargate** orchestration and **GitHub Actions** CI/CD, with **AWS CDK (TypeScript)**-defined infrastructure to allow contribution from non-familiar engineers
- Achieved a **34% speedup** on a pipeline routing **6,000+** scraped research documents for inference by implementing **Kinesis** streams feeding **Lambda** processors handling data processing and redirection, with **Lambda** + **API Gateway** fronting for app integration
- Integrated automated fine-tuning of an **AWS Bedrock** source relevancy inference model through **AWS EventBridge** cron triggers with decision gates to avoid redundant retraining, with **CloudWatch** + **X-Ray** application tracing for automated performance monitoring and metric-based **SNS** notifications to engineers on application failure

Machine Learning Research Intern

University of Maryland Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

College Park, MD

- Developing automated training for **PhysTwin**, a physics-based digital twin, with a **PhD research team** targeting an **ICLR** submission
- Built a **Python** pipeline linking **AnyGrasp** multi-camera grasp candidates to **Open3D**-rendered overlay views consumed by a **Qwen3-VL-8B-Instruct-FP8 VLM**, enabling researchers to develop changes faster through isolating scene components
- Cut research-environment setup time **30%** via **UV** dependency pinning and **Docker** for ensuring identical setup, with **psutil**-based timing and memory monitoring across runtime and **Pytest** unit testing for quickly identifying bottlenecks

Co-Founder, Software Engineer

TerpLabs

June 2025 – Present

College Park, MD

- Co-founded and scaled a **110+ member** campus product studio shipping **5** live products, utilizing **Jira** task management and **GitHub** code management to organize teams through agile practices
- Sustained **20,000+** site visits within the initial 30-day launch period of Tortuga, an enhanced class scheduler, through developing **Prisma**-based **SQL** querying with **Redis** query caching to reduce database load and application latency
- Building a work-in-progress integration of a **RAG** meal suggestion into the **official UMD app** using **GPT-4o mini** over **ChromaDB** + **SentenceBERT** vector search on scraped recipe data, achieving **93%** generated meal satisfaction on internal test users

PROJECTS

IEX Matching Engine Simulation | C++, STL, POSIX, Database internals

July 2026 – August 2026

- Implemented a **C++**-based order book simulation on historical IEX data, with a contract-based plug-and-play design to allow for evaluating multiple approaches with enforced correctness
- Wrote and evaluated paginated hash-based data storage for live order book data with in-memory **LFU** caching for frequently requested tickers during simulation, evaluated for drift against IEX's historical book data
- Designed and assessed the performance of a time-based **Adaptive LSM** storage against a **B+ tree** storage for logs during simulation, testing for throughput and latency via **rdtscp**-based timing and hardware utilization via Linux's **perf** performance counter

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript/TypeScript, HTML, CSS, Java, Rust, CUDA

Web & App Development: REST APIs (Express.js, Flask, FastAPI), Node.js, React.js/Next.js, Spring Boot

Data & Machine Learning: SQL, MongoDB, Pandas, NumPy, Scikit-learn, PyTorch

Cloud Development: Compute (EC2, Lambda, ALB, ECS, EKS), CI/CD (CloudFormation, CDK, CodeDeploy)

Cloud Security & Monitoring: Security (IAM, VPC), Observability (CloudWatch, SQS, SNS, X-Ray)

AI engineering: LLM tooling (Claude, Cursor, Copilot), HuggingFace, Vector-based RAG (Pinecone, FAISS, LangChain, LlamaIndex)

Tooling: Docker, Kubernetes, Version Control (Git/GitHub, Bitbucket), Linux, Unit Testing (Pytest, Jest)